

2021-Ph-H1-Q4

Section: Our Dynamic Universe

Topic: Forces

A 60 kg object is pulled up a slope with a 500 N force.
Find the resulting acceleration.

Solution:

- Weight component down slope:

$$mg \sin \theta = 60 \times 9.8 \times \sin 40^\circ \approx 378 \text{ N}$$

- Resultant force:

$$F_R = 500 - 378 = 122 \text{ N}$$

- Acceleration:

$$a = \frac{F}{m} = \frac{122}{60} \approx 2.0 \text{ m/s}^2$$

Answer: A

Guidance for Students:

Always resolve weight **parallel to the slope**. Use Newton's second law after finding net force.

Revision Tip:

Draw a diagram! And remember:

$$F = ma \quad \text{and} \quad \text{down - slope force} = mg \sin \theta$$